

An idea / mechanism note on per-generation chiral consistency from combined
Ward, Einstein–Hilbert, Witten, and mod-2 constraints in TECT
(*IDEA / MECHANISM NOTE; NOT a current theorem paper. External use
FORBIDDEN at the present canonical tier per Math314-AddC audit. Full
theorem-level rewrite tracked under Q-2026-05-02-Pillar7-Three-Generation-Rewrite.*)

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This document is an idea / mechanism note outlining a candidate framework for per-generation chiral consistency in the Topological Energy Condensate Theory (TECT) combining four independent consistency audits: (1) global Ward identities at each generation (Math47); (2) preservation of those Ward currents under nonlinear coupling to a fluctuating background metric (Math48); (3) the Witten $SU(2)_L$ global anomaly mod-2 constraint (Math49b-v3, after [1]); and (4) the mod-2 spectral-flow constraint on the chiral-fermion Dirac operator (Math49c-v3). *Honest scope and audit-flag (Math314-AddC)*. The present note does *not* establish that the four conditions together force the SM three-generation count: each is a per-generation consistency *requirement*, not a uniqueness theorem. The generation-count question — “why exactly three?” — remains the canonical Pillar 7 **T1 open** problem in TECT. The correction history is recorded in the file header (Rev 4, 2026-05-06). The companion paper Paper-07-ext records the symbolic-exact cancellation of the five SM perturbative gauge anomalies and the $\text{Tr}(U(1)_Y)$ gravitational mixed anomaly per generation; that result is independent of N_g and is the canonical $SO(10)$ anomaly-cancellation audit of the TECT Pillar 4 embedding. External use of this note is FORBIDDEN at the present canonical tier; a full theorem-level rewrite of any three-generation-forcing claim is tracked under Q-2026-05-02-Pillar7-Three-Generation-Rewrite.

I. INTRODUCTION

The existence of exactly three SM fermion generations is one of the deepest empirical puzzles of the Standard Model. No first-principles derivation has been universally accepted; the count $N_g = 3$ remains a phenomenological input. TECT formulates the question in a constructive form — can the topological-condensate vacuum be required, by some combination of field-theoretic consistency

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conditions, to admit only $N_g = 3$? — and answers it, at the present canonical tier, in the *negative*: the per-generation consistency audits identified here are necessary, but they are not collectively known to be sufficient.

The present mechanism note assembles the four audit conditions that, in combination, are the natural starting point for any future three-generation-forcing argument inside TECT. They are: the global Ward identity per generation (§III), the preservation of that Ward current under nonlinear coupling to a fluctuating background metric (§IV), the Witten $SU(2)_L$ global anomaly mod-2 constraint (§V), and the mod-2 spectral-flow constraint on the chiral-fermion Dirac operator (§VI). None of them, individually or jointly, is known to force $N_g = 3$ at the level of the present construction.

The paper is therefore positioned as an idea / mechanism note on *which audits* a future theorem-level construction must pass; it is not, and does not claim to be, the theorem-level construction itself.

II. SETTING AND MAIN CLAIM (RECAST AS CONJECTURE)

For each generation g of the SM, define the chiral fermion content in the standard $SU(3)_c \times SU(2)_L \times U(1)_Y$ representation:

$$Q_L^g \in (\mathbf{3}, \mathbf{2})_{1/6}, \quad u_R^{c,g} \in (\overline{\mathbf{3}}, \mathbf{1})_{-2/3}, \quad d_R^{c,g} \in (\overline{\mathbf{3}}, \mathbf{1})_{1/3}, \quad L_L^g \in (\mathbf{1}, \mathbf{2})_{-1/2}, \quad e_R^{c,g} \in (\mathbf{1}, \mathbf{1})_1, \quad (1)$$

together with the right-handed neutrino singlet $\nu_R^{c,g} \in (\mathbf{1}, \mathbf{1})_0$ (the RHN; SM hypercharge $Y = 0$) when $SO(10)$ is the embedding GUT. The global Ward current of generation g for any conserved $U(1)$ flavour symmetry X is $j_{X,g}^\mu = \overline{\Psi}^g \gamma^\mu X \Psi^g$, satisfying $\partial_\mu j_{X,g}^\mu = 0$ at the classical level; quantum corrections modify this only via gauge anomalies, which are audited in the companion paper Paper-07-ext.

Conjecture 1 (Per-generation quantum-consistency four-condition audit; **T2 conjecture**). For each generation $g \in \{1, \dots, N_g\}$ of the SM matter content (1), the following four conditions are *necessary* for quantum consistency of the per-generation chiral sector:

- (**Ward-g**) global Ward identity per Math47 (see §III): $\partial_\mu j_{X,g}^\mu = 0$ for every classically conserved global $U(1)_X$ symmetry, modulo the gauge anomalies that cancel between generations as audited in Paper-07-ext;
- (**EH-g**) preservation of the Ward current $j_{X,g}^\mu$ under nonlinear coupling to a fluctuating background metric, per the one-loop Math48 audit (§IV);

(**Witten-tot**) the *total* chiral $SU(2)_L$ doublet count of the theory is even, per the Witten $SU(2)$ global anomaly (Math49b-v3 + [1]; §V);

(**Mod-2-g**) the mod-2 spectral-flow index of the chiral Dirac operator on the BCC shell, per the Math49c-v3 audit (§VI), is even for each generation.

The four-condition conjunction $(\text{Ward-}g) \wedge (\text{EH-}g) \wedge (\text{Witten-tot}) \wedge (\text{Mod-2-}g)$ is necessary for the per-generation chiral sector to admit a well-defined quantum extension.

Honest scope. The conjunction is *not* known to be *sufficient* to force the value of N_g . The generation-count question — “does the four-condition conjunction admit solutions only at $N_g = 3$?” — is the canonical Pillar 7 **T1 open** problem and is the subject of the rewrite obligation Q-2026-05-02-Pillar7-Three-Generation-Rewrite.

Remark 2 (On the difference from the earlier draft). Earlier drafts of this paper asserted that the conjunction *forces* $N_g = 3$ at **T7 proved**. That assertion is withdrawn in Rev 4 because (i) the Witten $SU(2)$ global anomaly is even for $N_g \in \{1, 2, 3, \dots\}$ (the per-generation doublet count is 4, always even), and therefore does not discriminate $N_g = 3$; (ii) the formula $A_{\text{grav}} = \sum_f Q_f^3$ used in the earlier “Step 2” is *not* the SM gravitational anomaly coefficient (the actual mixed gravitational anomaly is $\text{Tr}(U(1)_Y)$, with one power of Y , which vanishes per generation independent of N_g ; this is the canonical $SO(10)$ audit of Paper-07-ext); (iii) the mod-2 parity expression $(-1)^{N_g(N_g-1)/2}$ in the earlier “Step 4” is non-standard and not tied to any canonical TECT construction. The corrected statement is recorded in Conjecture 1.

III. AUDIT 1 — PER-GENERATION WARD IDENTITY (MATH47)

Math47 establishes that, for the SM coupled to the TECT background condensate, the global Ward identity for any classically conserved $U(1)_X$ flavour current of generation g is preserved at all loop orders, provided the gauge-anomaly cancellation conditions of Paper-07-ext are satisfied for the full chiral spectrum. The precise theorem is the per-generation independence statement: the Ward current $j_{X,g}^\mu$ does not mix with $j_{X,g'}^\mu$ for $g \neq g'$ at tree level, and the loop-level mixing is governed by the same gauge-anomaly trace formulas audited in Paper-07-ext.

Remark 3 (What Math47 does *not* establish). Math47 does not by itself force any particular value of N_g : the per-generation Ward identity holds at every g *individually*, and the gauge-anomaly cancellation it depends on is, per the six-coefficient table (eq. 6 of Paper-07-ext), N_g -independent.

IV. AUDIT 2 — PRESERVATION UNDER NONLINEAR EH COUPLING (MATH48)

When the SM matter is coupled to a fluctuating background metric in the linearised regime $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, the mixed gravitational anomaly potentially obstructs the conservation of each Ward current. Math48 establishes, at one-loop, the nonlinear preservation of $\partial_\mu j_{X,g}^\mu = 0$ provided the gravitational mixed anomaly $\text{Tr}(\text{U}(1)_X)$ vanishes for every gauged X . For $X = Y$ (the SM hypercharge), this condition is the audit-target of Paper-07-ext, where it is proven symbolically exact zero per generation (2); for $X = B - L$ (the $\text{SO}(10)$ extension), it is closed by the RHN.

Remark 4 (Removal of the non-standard $A_{\text{grav}} = \sum_f Q_f^3$ formula). The earlier draft (Rev 3) introduced the formula $A_{\text{grav}} = \sum_f Q_f^3$ (sum over fermions of the cube of the electric charge Q) as the “gravitational anomaly coefficient”. This is incorrect: the gravitational mixed anomaly in four dimensions is the trace $\text{Tr}(\text{U}(1)_X)$ for the gauged Abelian charge X (one power, not three), arising from the graviton–graviton–gauge-boson triangle. The pure-gauge anomaly $\text{Tr}(\text{U}(1)_X^3)$ (three powers) is a *separate* coefficient, and for $X = Q$ (electric charge) is the non-canonical combination of $\text{U}(1)_Y$ and $\text{SU}(2)_L$ contributions which is normally evaluated only in the broken electroweak phase below the Higgs vev. The earlier draft conflated these two coefficients. In Rev 4 the formula is removed and the audit is the standard $\text{Tr}(\text{U}(1)_Y)$ as recorded in Paper-07-ext, equation (2).

A local stub of the canonical identity, proved as in the six-coefficient table of the companion paper Paper-07-ext, is recorded here for self-contained citation:

$$\text{Tr}(\text{U}(1)_Y) = 0 \quad \text{per generation} \quad [\text{symbolic identity, proven in Paper-07-ext}]. \quad (2)$$

V. AUDIT 3 — WITTEN $\text{SU}(2)_L$ GLOBAL ANOMALY (MATH49B-V3, AFTER WITTEN 1982)

Witten [1] proved that a four-dimensional chiral gauge theory with gauge group $\text{SU}(2)$ is consistent at the non-perturbative level only if the total number of chiral $\text{SU}(2)$ doublets is even. The argument uses $\pi_4(\text{SU}(2)) = \mathbb{Z}_2$: a homotopically nontrivial $\text{SU}(2)$ gauge transformation generates a sign in the fermionic path-integral measure that vanishes only if the total doublet count is even.

For the SM at N_g generations, the chiral $\text{SU}(2)_L$ doublets per generation are: the three quark doublets (one per colour) of Q_L , plus the lepton doublet L_L . The total per-generation doublet count is therefore $3 + 1 = 4$, which is even. The total over N_g generations is $4N_g$, which is even for every positive integer N_g .

Remark 5 (Witten does not discriminate $N_g = 3$). The Witten $SU(2)_L$ global anomaly is satisfied for every $N_g \geq 1$. It does not discriminate $N_g = 1$ from $N_g = 3$. The earlier draft assertion that “the coefficient is zero iff $N_g = 3$ ” is corrected in Rev 4: the coefficient (the $\mathbb{Z}_2$ class) is zero for every N_g .

VI. AUDIT 4 — MOD-2 SPECTRAL FLOW ON THE BCC SHELL (MATH49C-V3)

Math49c-v3 examines the chiral Dirac operator on the BCC shell of the TECT condensate vacuum. Its integer-valued Atiyah–Singer index, per generation, is given by a topological-charge sum on the shell; quantum consistency of the chiral path integral requires the mod-2 reduction of this index to be even (so that the fermionic Pfaffian is well-defined as a real-valued square root of the chiral Dirac determinant).

Remark 6 (Mod-2 does not, in the present construction, force $N_g = 3$). The mod-2 spectral-flow index of Math49c-v3 is, in the present canonical construction, an integer that is even for the SM $SU(5)$ content of $\bar{\mathbf{5}} \oplus \mathbf{10}$ per generation; this is consistent with $N_g = 1, 2, 3, \dots$ alike. A construction in which the index parity is sensitive to N_g is conceivable but is not part of the present Math49c-v3 archive; if completed, it would be the central technical step of any future three-generation-forcing theorem.

VII. CLOSURE AND HONEST VERDICT

The four audits above are individually closed at **T6 proved conditional** or stronger (Ward, EH, Witten, mod-2 each rest on textbook or Math47/Math48/Math49b-v3/Math49c-v3 canonical archive). *Their conjunction is necessary* for the per-generation chiral sector to admit a well-defined quantum extension.

The conjunction is not known to be sufficient to force the SM generation count. Specifically:

- Witten $SU(2)_L$ is satisfied for every $N_g \geq 1$;
- Mod-2 spectral flow, in the present Math49c-v3 construction, is satisfied for every N_g for which the chiral $SU(5)$ content sums to an even index — which is the case for every N_g ;
- Ward and EH are per-generation conditions that hold individually for each $g \in \{1, \dots, N_g\}$ and are N_g -independent;

- the perturbative SM gauge anomalies and $\text{Tr}(U(1)_Y)$ gravitational anomaly cancel *per generation* (Paper-07-ext, equation (2) and the six-coefficient table of that paper), again N_g -independent.

The honest verdict is therefore that the four-condition consistency audit, combined with the Paper-07-ext anomaly cancellation, fixes N_g to be a positive integer (each generation must satisfy the audit) but *does not* fix $N_g = 3$. The phenomenological value $N_g = 3$ remains the empirical input, and the Pillar 7 question “why exactly three?” is canonical **T1 open** in TECT, tracked under **Q-2026-05-02-Pillar7-Three-Generation-Rewrite**.

VIII. DISCUSSION AND OUTLOOK

The mechanism note recorded here serves three purposes within the TECT Pillar 7 programme:

1. to enumerate the necessary per-generation consistency audits that any future three-generation-forcing theorem must pass;
2. to record the Math314-AddC corrections to the earlier over-claims (Rev 3 had asserted $N_g = 3$ forced at **T7 proved**, an over-claim now retracted);
3. to provide a clean target for the **Q-2026-05-02-Pillar7-Three-Generation-Rewrite** obligation, which would supply a fifth condition (a TECT-internal construction sensitive to N_g) sufficient to force $N_g = 3$ in conjunction with the four conditions above.

The fifth condition, if ever constructed, would most likely arise from the BCC shell topology — the integer-valued index of the Math49c-v3 chiral Dirac operator restricted to the three-band shell spectrum (Math234), whose three-fold degeneracy might be tied to a TECT-internal Chern number or flux quantum. Such a construction is a canonical Pillar 7 T1 OPEN item and is not part of the present note.

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- [1] E. Witten, “An $SU(2)$ anomaly,” *Phys. Lett. B* **117**, 324 (1982).
 [2] TECT internal note Math47, “Per-generation Ward identity in the SM coupled to TECT condensate,” TECT canonical archive.
 [3] TECT internal note Math48, “Nonlinear EH coupling and Ward-current preservation at one-loop,” TECT canonical archive.

- [4] TECT internal note Math49b-v3, “Witten $SU(2)_L$ global anomaly audit for the TECT chiral spectrum,” TECT canonical archive.
- [5] TECT internal note Math49c-v3, “Mod-2 spectral flow on the BCC shell chiral Dirac operator,” TECT canonical archive.
- [6] TECT internal note Math314-AddC, “Hostile-referee audit of Wave-4/5 papers Paper-06..08; rewrite obligation Q-2026-05-02-Pillar7-Three-Generation-Rewrite,” TECT canonical archive.

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